

Multiple Choice (10 marks)

1. Which of the following genes conveys susceptibility for polygenic Alzheimer disease?
 - (a) APP
 - (b) PS 1
 - (c) PS 2
 - (d) APOE
2. The normal human chromosome diploid number is:
 - (a) 23
 - (b) 24
 - (c) 46
 - (d) 48
3. Most viral vaccines are produced by which of the following techniques?
 - (a) Traditional cell culture
 - (b) Molecular biology
 - (c) Animal tissues in vitro
 - (d) VLPs (virus like particles)
4. The highest risk of HIV infection among drug users in the U.S. is in:
 - (a) The east coast
 - (b) The central United States (Kansas, Missouri etc.)
 - (c) Southern California
 - (d) The Pacific Northwest (Washington, Oregon, Idaho)
5. Free radicals are produced by
 - (a) Normal metabolism
 - (b) Excess consumption of Vitamins A and C
 - (c) Lack of sunlight
 - (d) A weakened immune system

True / False (10 marks)

Answer True or False

6. True or False: The dorsal roots of all spinal nerves contain both sensory and motor neuronal processes that facilitate reflex actions.
7. True or False: The gender gap is likely associated with differences in hormones, risky behaviors, and smoking prevalence between males and females.
8. True or False: The trachea is totally enclosed by cartilaginous rings throughout its length.
9. True or False: During palpation of the neck, the cricoid cartilage should be felt before the thyroid cartilage and hyoid bone.
10. True or False: As of 2019, African American women are more likely to remain at home alone compared to other racial and ethnic groups.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of tar in tobacco smoke and its contribution to the development of cancers. Analyze its mechanisms and the implications for public health.
12. Discuss the role of cellular scaffolding in the nucleus and cytoplasm during the assembly stage of viral replication, highlighting its significance and mechanism.

End of Paper